

### SAFETY DATA SHEET

Revision Date 01-Jul-2014

Revision Number 1

#### 1. Identification

**Product Name** R21226 Gelatin Reagent

**Cat No. :** R21226

**Synonyms** No information available

**Recommended Use** Laboratory chemicals.

**Uses advised against** No Information available

**Details of the supplier of the safety data sheet**

**Emergency Telephone Number**

Chemtrec US: (800) 424-9300

Chemtrec EU: 001 (202) 483-7616

#### 2. Hazard(s) identification

##### Classification

Classification under 2012 OSHA Hazard Communication Standard (29 CFR 1910.1200)

Based on available data, the classification criteria are not met

##### Label Elements

None required

##### Hazards not otherwise classified (HNOC)

None identified

##### Unknown Acute Toxicity

.? % of the mixture consists of ingredients of unknown toxicity.

#### 3. Composition / information on ingredients

| Component         | CAS-No    | Weight % |
|-------------------|-----------|----------|
| Water             | 7732-18-5 | 73.96    |
| Mercuric chloride | 7487-94-7 | 11.11    |
| Hydrogen chloride | 7647-01-0 | 14.93    |

#### 4. First-aid measures

|                                        |                                                                                          |
|----------------------------------------|------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes.                       |
| <b>Inhalation</b>                      | Move to fresh air.                                                                       |
| <b>Ingestion</b>                       | Do not induce vomiting.                                                                  |
| <b>Most important symptoms/effects</b> | No information available.                                                                |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                    |

## 5. Fire-fighting measures

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Unsuitable Extinguishing Media</b>   | No information available |
| <b>Flash Point</b>                      | No information available |
| <b>Method -</b>                         | No information available |
| <b>Autoignition Temperature</b>         | No information available |
| <b>Explosion Limits</b>                 |                          |
| <b>Upper</b>                            | No data available        |
| <b>Lower</b>                            | No data available        |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

### Specific Hazards Arising from the Chemical

Keep product and empty container away from heat and sources of ignition.

### Hazardous Combustion Products

None known

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
0

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

|                                  |                                              |
|----------------------------------|----------------------------------------------|
| <b>Personal Precautions</b>      | Ensure adequate ventilation.                 |
| <b>Environmental Precautions</b> | Should not be released into the environment. |

**Methods for Containment and Clean Up** No information available.

## 7. Handling and storage

|                 |                                                                          |
|-----------------|--------------------------------------------------------------------------|
| <b>Handling</b> | Ensure adequate ventilation.                                             |
| <b>Storage</b>  | Keep containers tightly closed in a dry, cool and well-ventilated place. |

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component         | ACGIH TLV                            | OSHA PEL                                                                                                             | NIOSH IDLH                                                                                  |
|-------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Mercuric chloride | TWA: 0.025 mg/m <sup>3</sup><br>Skin | (Vacated) Ceiling: 0.1 mg/m <sup>3</sup>                                                                             | IDLH: 10 mg/m <sup>3</sup><br>TWA: 0.05 mg/m <sup>3</sup><br>Ceiling: 0.1 mg/m <sup>3</sup> |
| Hydrogen chloride | Ceiling: 2 ppm                       | Ceiling: 5 ppm<br>Ceiling: 7 mg/m <sup>3</sup><br>(Vacated) Ceiling: 5 ppm<br>(Vacated) Ceiling: 7 mg/m <sup>3</sup> | IDLH: 50 ppm<br>Ceiling: 5 ppm<br>Ceiling: 7 mg/m <sup>3</sup>                              |

| Component         | Quebec                                           | Mexico OEL (TWA)                               | Ontario TWAEV                        |
|-------------------|--------------------------------------------------|------------------------------------------------|--------------------------------------|
| Mercuric chloride | TWA: 0.025 mg/m <sup>3</sup><br>Skin             | TWA: 0.05 mg/m <sup>3</sup>                    | TWA: 0.025 mg/m <sup>3</sup><br>Skin |
| Hydrogen chloride | Ceiling: 5 ppm<br>Ceiling: 7.5 mg/m <sup>3</sup> | Ceiling: 5 ppm<br>Ceiling: 7 mg/m <sup>3</sup> | CEV: 2 ppm                           |

Legend

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH IDLH: The National Institute for Occupational Safety and Health Immediately Dangerous to Life or Health

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas.

**Personal Protective Equipment****Eyel/face Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

**Respiratory Protection**

Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                          |
|----------------------------------------|--------------------------|
| Physical State                         | Liquid                   |
| Appearance                             | No information available |
| Odor                                   | No information available |
| Odor Threshold                         | No information available |
| pH                                     | No information available |
| Melting Point/Range                    | No data available        |
| Boiling Point/Range                    | No information available |
| Flash Point                            | No information available |
| Evaporation Rate                       | No information available |
| Flammability (solid,gas)               | No information available |
| Flammability or explosive limits       |                          |
| Upper                                  | No data available        |
| Lower                                  | No data available        |
| Vapor Pressure                         | No information available |
| Vapor Density                          | No information available |
| Relative Density                       | No information available |
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | No information available |
| Decomposition temperature              | No information available |
| Viscosity                              | No information available |

## 10. Stability and reactivity

**Reactive Hazard**

None known, based on information available

|                                         |                                              |
|-----------------------------------------|----------------------------------------------|
| <b>Stability</b>                        | Stable under recommended storage conditions. |
| <b>Conditions to Avoid</b>              | Incompatible products.                       |
| <b>Incompatible Materials</b>           | Strong oxidizing agents                      |
| <b>Hazardous Decomposition Products</b> | None under normal use conditions             |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.     |
| <b>Hazardous Reactions</b>              | None under normal processing.                |

## 11. Toxicological information

### Acute Toxicity

|                    |                                                                               |
|--------------------|-------------------------------------------------------------------------------|
| <b>Oral LD50</b>   | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| <b>Dermal LD50</b> | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| <b>Mist LC50</b>   | Based on ATE data, the classification criteria are not met. ATE > 5 mg/l.     |
| <b>Vapor LC50</b>  | Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.    |

### **Component Information** **Toxicologically Synergistic** **Products**

No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** This product contains one or more substances which are classified by IARC as carcinogenic to humans (Group I), probably carcinogenic to humans (Group 2A) or possibly carcinogenic to humans (Group 2B).

| Component         | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-------------------|-----------|------------|------------|------------|------------|------------|
| Water             | 7732-18-5 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Mercuric chloride | 7487-94-7 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Hydrogen chloride | 7647-01-0 | group 3    | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known  
**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains.

| Component         | Freshwater Algae | Freshwater Fish                                                                                                                                                                                                                                                                      | Microtox   | Water Flea                                   |
|-------------------|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------------------------------------------|
| Mercuric chloride | Not listed       | 5.933 - 10.34 mg/L LC50 96 h<br>0.041 mg/L LC50 96 h 0.1<br>- 0.182 mg/L LC50 96 h<br>0.155 mg/L LC50 96 h 0.13<br>- 0.19 mg/L LC50 96 h 0.014<br>- 0.019 mg/L LC50 96 h 0.02<br>- 0.26 mg/L LC50 96 h 4.425<br>mg/L LC50 96 h 0.4 mg/L<br>LC50 96 h 0.096 - 0.133<br>mg/L LC50 96 h | Not listed | EC50=0.0015mg/L 48 h<br>EC50=0.012mg/L >48 h |
| Hydrogen chloride | Not listed       | 282 mg/L LC50 96 h                                                                                                                                                                                                                                                                   | Not listed | Not listed                                   |

**Persistence and Degradability** No information available

**Bioaccumulation/ Accumulation** No information available.

**Mobility** No information available.

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

**DOT** Not regulated  
**TDG** Not regulated  
**IATA** Not regulated  
**IMDG/IMO** Not regulated

### 15. Regulatory information

All of the components in the product are on the following Inventory lists: Australia X = listed China Canada Europe TSCA Korea Philippines

#### International Inventories

| Component         | TSCA | DSL | NDSL | EINECS    | ELINCS | NLP | PICCS | ENCS | AICS | IECSC | KECL |
|-------------------|------|-----|------|-----------|--------|-----|-------|------|------|-------|------|
| Water             | X    | X   | -    | 231-791-2 | -      |     | X     | -    | X    | X     | X    |
| Mercuric chloride | X    | X   | -    | 231-299-8 | -      |     | X     | X    | X    | X     | X    |
| Hydrogen chloride | X    | X   | -    | 231-595-7 | -      |     | X     | X    | X    | X     | X    |

**Legend:**

X - Listed

E - Indicates a substance that is the subject of a Section 5(e) Consent order under TSCA.

F - Indicates a substance that is the subject of a Section 5(f) Rule under TSCA.

N - Indicates a polymeric substance containing no free-radical initiator in its inventory name but is considered to cover the designated polymer made with any free-radical initiator regardless of the amount used.

P - Indicates a commenced PMN substance

R - Indicates a substance that is the subject of a Section 6 risk management rule under TSCA.

S - Indicates a substance that is identified in a proposed or final Significant New Use Rule

T - Indicates a substance that is the subject of a Section 4 test rule under TSCA.

XU - Indicates a substance exempt from reporting under the Inventory Update Rule, i.e. Partial Updating of the TSCA Inventory Data Base Production and Site Reports (40 CFR 710(B)).

Y1 - Indicates an exempt polymer that has a number-average molecular weight of 1,000 or greater.

Y2 - Indicates an exempt polymer that is a polyester and is made only from reactants included in a specified list of low concern reactants that comprises one of the eligibility criteria for the exemption rule.

#### U.S. Federal Regulations

**TSCA 12(b)** Not applicable

**SARA 313** Not applicable

| Component         | CAS-No    | Weight % | SARA 313 - Threshold Values % |
|-------------------|-----------|----------|-------------------------------|
| Mercuric chloride | 7487-94-7 | 11.11    | 1.0                           |
| Hydrogen chloride | 7647-01-0 | 14.93    | 1.0                           |

**SARA 311/312 Hazardous Categorization**

|                                   |    |
|-----------------------------------|----|
| Acute Health Hazard               | No |
| Chronic Health Hazard             | No |
| Fire Hazard                       | No |
| Sudden Release of Pressure Hazard | No |
| Reactive Hazard                   | No |

Clean Water Act Not applicable

| Component         | CWA - Hazardous Substances | CWA - Reportable Quantities | CWA - Toxic Pollutants | CWA - Priority Pollutants |
|-------------------|----------------------------|-----------------------------|------------------------|---------------------------|
| Mercuric chloride | -                          | -                           | X                      | -                         |
| Hydrogen chloride | X                          | 5000 lb                     | -                      | -                         |

Clean Air Act Not applicable

| Component         | HAPS Data | Class 1 Ozone Depletors | Class 2 Ozone Depletors |
|-------------------|-----------|-------------------------|-------------------------|
| Mercuric chloride | X         |                         | -                       |
| Hydrogen chloride | X         |                         | -                       |

**OSHA Occupational Safety and Health Administration**

Not applicable

| Component         | Specifically Regulated Chemicals | Highly Hazardous Chemicals |
|-------------------|----------------------------------|----------------------------|
| Hydrogen chloride | -                                | TQ: 5000 lb                |

**CERCLA**

Not applicable

| Component         | Hazardous Substances RQs | CERCLA EHS RQs |
|-------------------|--------------------------|----------------|
| Mercuric chloride | -                        | 500 lb         |
| Hydrogen chloride | 5000 lb                  | 5000 lb        |

California Proposition 65 This product does not contain any Proposition 65 chemicals

| Component         | CAS-No    | California Prop. 65 | Prop 65 NSRL | Category      |
|-------------------|-----------|---------------------|--------------|---------------|
| Mercuric chloride | 7487-94-7 | Developmental       | -            | Developmental |

State Right-to-Know Not applicable

| Component         | Massachusetts | New Jersey | Pennsylvania | Illinois | Rhode Island |
|-------------------|---------------|------------|--------------|----------|--------------|
| Mercuric chloride | X             | X          | X            | X        | -            |
| Hydrogen chloride | X             | X          | X            | X        | X            |

**U.S. Department of Transportation**

|                             |   |
|-----------------------------|---|
| Reportable Quantity (RQ):   | N |
| DOT Marine Pollutant        | N |
| DOT Severe Marine Pollutant | N |

**U.S. Department of Homeland Security**

This product does not contain any DHS chemicals.

| Component         | DHS Chemical Facility Anti-Terrorism Standard                     |
|-------------------|-------------------------------------------------------------------|
| Hydrogen chloride | 0 lb STQ (anhydrous); 11250 lb STQ (37% concentration or greater) |

**Other International Regulations**

Mexico - Grade No information available

**Canada**

This product has been classified in accordance with the hazard criteria of the Controlled Products Regulations (CPR) and the MSDS contains all the information required by the CPR

**WHMIS Hazard Class**

Non-controlled

**16. Other information****Prepared By**

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Thermo Fisher Scientific  
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**Revision Date**

01-Jul-2014

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01-Jul-2014

**Revision Summary**

This document has been updated to comply with the US OSHA HazCom 2012 Standard replacing the current legislation under 29 CFR 1910.1200 to align with the Globally Harmonized System of Classification and Labeling of Chemicals (GHS)

**Disclaimer**

The information provided on this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guide for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered as a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other material or in any process, unless specified in the text.

**End of SDS**